CAI 6108 – Machine Learning Engineering:

Explainable/Interpretable ML

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Spring 2026

Administrivia

- Homework 5
  - Topic: CNNs, RNNs/Transformers, AutoEncoders
  - This is the **last homework**; after that focus on the project and final exam
  - Due 4/17

- Final Project Deliverables
  - Written **report** and **code** due 4/22 (no penalty if submitted by 4/24)
    - See Canvas assignment for additional instructions and template (LaTex)

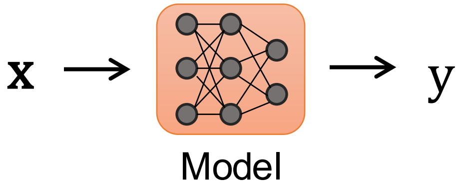
Administrivia: Final Exam

- Final Exam is coming up!
  - **Wednesday 4/22** during class time (10:40 - 11:30) in **NRN 1020**
    - Topics: everything until 4/20 but focus on topics covered after the midterm
  - Duration: 50 minutes
  - Format: **closed-books**, (blank) scratch paper, and physical calculator are allowed (no phones!)
    - ***Same as the midterm***
  - Questions: short answers (may include MCQs) + problems

Administrivia

- **Course Evaluation**
  - Help us improve the course: complete evaluation by **April 24**
  - Access the evaluation form:
    - Canvas: click on GatorEvals (left navigation panel)
    - or: https://ufl.bluera.com/ufl/
  - Optional and anonymous

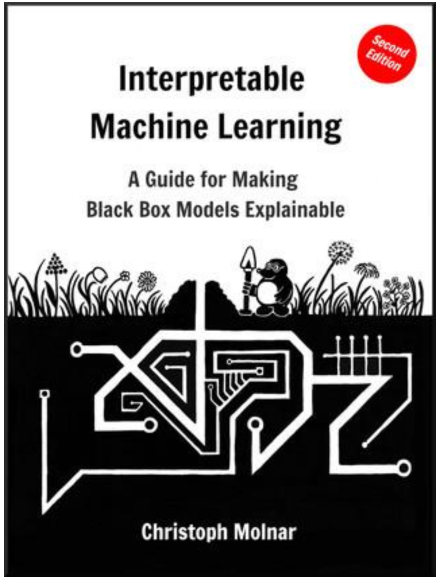
Reminder: Interpretable ML



- Most ML models are black-box in terms of their decisions
  - You feed an input x, you get an output y
  - Why did we get output y (and not some $y' \neq y$)?
- We want human-understandable explanations
  - How?

Reminder: Resources

- *"Interpretable Machine Learning: A Guide for Making Black Box Models Explainable."* Christoph Molnar.
  - Free HTML version of the book: https://christophm.github.io/interpretable-ml-book/

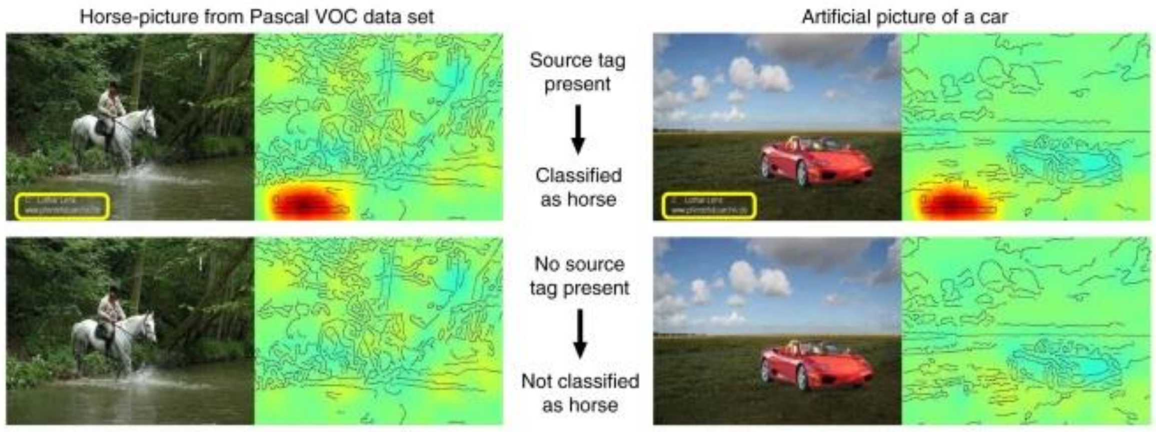


Reminder: Motivation

- Why do we need explanations?
  - For technology to be accepted and used, users must **trust** it (to some extent)
    - How much does the human understand the model's behavior?
  - Note: sometimes the stakes are high
    - E.g., some models are used for decision making (e.g., medical diagnosis, terrorism)
- Applications of interpretable ML
  - Validating models
    - E.g., fairness
  - Debugging models
    - E.g., adversarial examples
  - Human learning

Reminder: Do Not **Blindly Trust** Models

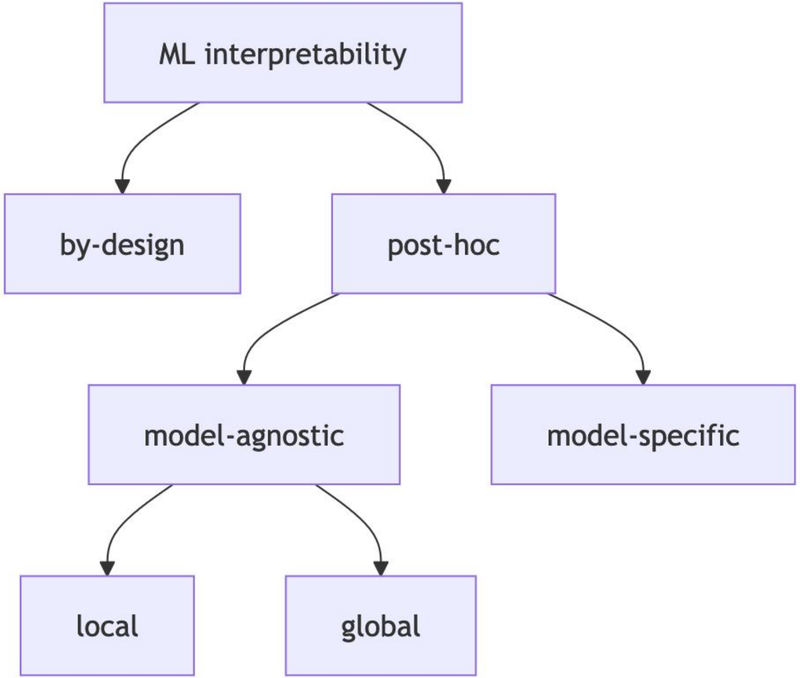
- Models can be correct for the wrong reasons!
  - The classifier focuses on a source tag present in 1/5 of horse pictures (PASCAL VOC 2007 dataset)
  - Example of spurious correlation in training data



Source: Lapuschkin et al. "Unmasking Clever Hans predictors and assessing what machines really learn." Nature communications 10.1, 2019.

Reminder: Taxonomy

- Taxonomy from Molnar 2025:



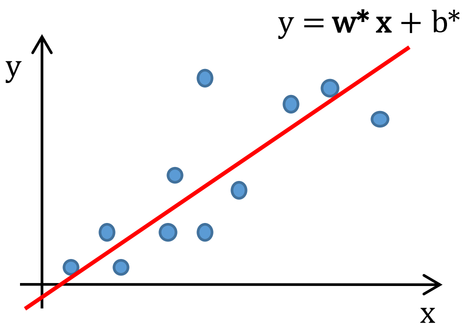
- By-design:
  - We train ***interpretable*** models
- Post-hoc:
  - Trained model not interpretable, but we use an **interpretability** method
- Model agnostic:
  - Some methods can be applied to **any model** type/architecture (E.g.: feature importance)
- Model specific:
  - Methods only applicable to **specific model types** (E.g.: attention networks)
- Global vs. local vs. glocal:
  - Local: explain a specific example
  - Global: explain entire datasets
  - Glocal: explain a subset of examples

Intepretability **By Design**

- What methods provide interpretable models?
  - Linear models, logistic regression, etc. Why?
    - Feature **importance/influence** on prediction is determined by **coefficients/weights**
    - However: beware of *multicollinearity*
  - Decision trees. Why?
    - The prediction for an example is "explained" by the ***path followed to a leaf***
    - If a tree is small, then we can easily visualize it and understand it fully
  - k-Nearest Neighbors. Why?
    - A prediction is "explained" by its k ***closest examples*** in the training data
  - Other methods (examples):
    - RuleFit: Friedman and Popescu. *"Predictive Learning via Rule Ensembles."* The Annals of Applied Statistics. 2008
    - ProtoViT: Ma et al. *"Interpretable image classification with adaptive prototype-based vision transformers."* NeurIPS 2024.

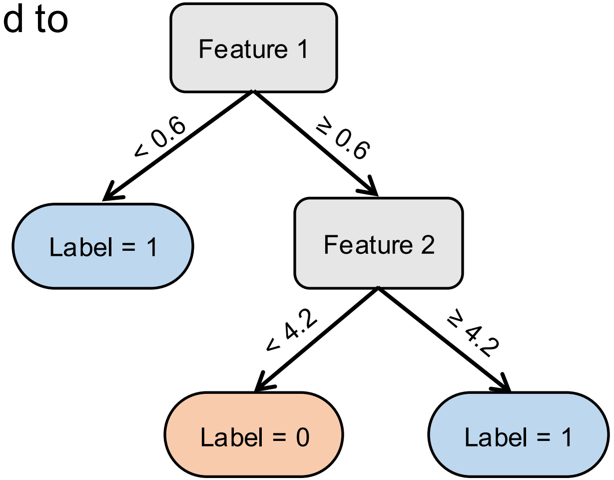
Reminder: Linear Regression

- Dataset
  - Matrix $\mathbf{X}$ $(n \times m)$ and the target vector $\mathbf{y}$ $(n \times 1)$
    - Let $\mathbf{x_i}$ be the feature vector for example i and $y_i \in R$ is the corresponding target/value
- Prediction task:
  - Given a feature vector $\mathbf{x}$, predict the target/value $y \in R$ as accurately as possible
- Linear Regression:
  - The model is: $h_\theta(\mathbf{x}) = h_{\mathbf{w},b}(\mathbf{x}) = \mathbf{w}\,\mathbf{x} + b$
  - The prediction is: $y = h_\theta(\mathbf{x})$
- Training:
  - We want to minimize the Mean Squared Error (MSE) [this is called OLS]
    - MSE($\mathbf{w}$,b) := MSE($h_{\mathbf{w},b}$, $\mathbf{X}$, $\mathbf{y}$) = $(1/n) \sum_i [h_{\mathbf{w},b}(\mathbf{x_i}) - y_i]^2 = (1/n) \sum_i [\mathbf{w}\,\mathbf{x_i} + b - y_i]^2$
  - Optimal parameters: $\boldsymbol{\theta}^* = (\mathbf{w}^*, b^*) = \text{argmin}_{\mathbf{w},b}$ MSE($\mathbf{w}$,b)



Reminder: Decision Trees

- A decision tree is
  - An *acyclic* graph (i.e., a directed rooted tree) that can be used to make predictions
  - Nonparametric model suitable for classification or regression
  - Prediction:
    - Start at the root
    - Traverse the tree (branching according to feature values)
    - The leaf gives the predicted label or value/target
  - How is the tree constructed from the training data?
    - There are many algorithms and many different kinds of decision trees!

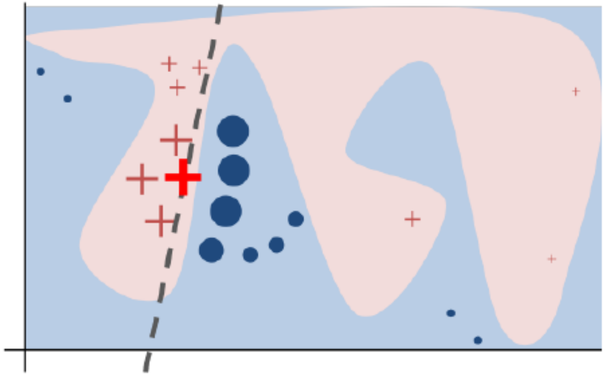


Post-hoc Interpretability

- Model agnostic:
  - Local:
    - LIME
    - SHAP / Shapley values
    - Others: Counterfactual Explanations, Individual Conditional Expectations (ICE), etc.
  - Global:
    - Permutation Feature Importance (PFI)
    - Leave One Feature Out Importance (LOFO)
    - Surrogate Models
    - Others: Prototypes & Criticisms, etc.
- Model specific (mostly for neural networks):
  - Rule extraction
  - Saliency maps
  - Concept activations

Interpretable ML: LIME

- Ribeiro, Marco Tulio, Sameer Singh, and Carlos Guestrin. "*Why should I trust you?: Explaining the predictions of any classifier*." KDD, 2016
  - Given an instance x
  - Idea: approximate the behavior of the model in a neighborhood of x using a **proxy model**
  - Choice for the proxy model:
    - Linear model, decision tree, falling rule list



- Decision function: blue-pink background
- Instance being explained: bold red cross
- Learned explanation: dashed line

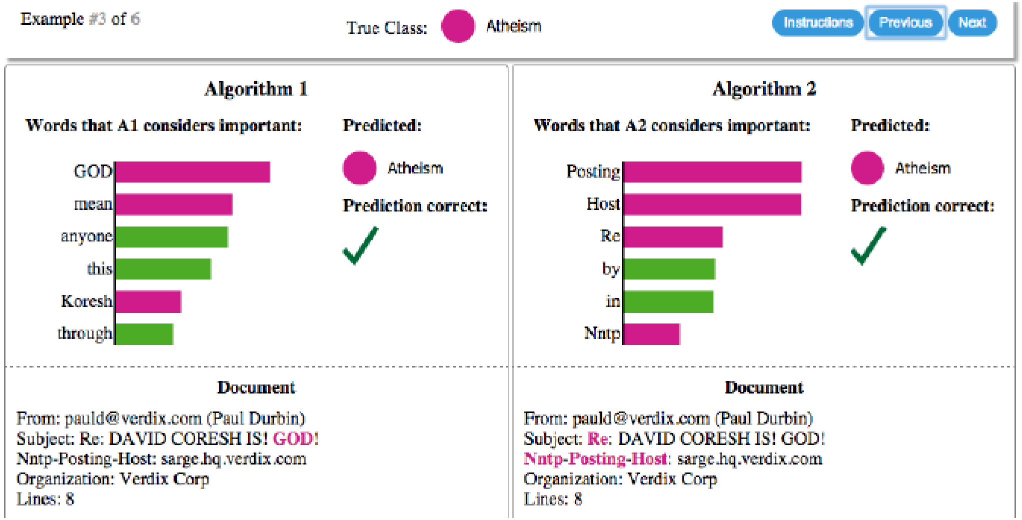
Interpretable ML: LIME

- Setup
  - f is the classifier
  - $\mathbf{x}$ is the input we want to explain ($f(\mathbf{x})=y$ for some y)
  - G is a class of interpretable models (e.g., linear models, decision trees, etc.)
    - $g \in G$ is an **explanation**
  - $\Omega(g)$ is a measure of complexity of explanation (e.g., depth of tree if g is a decision tree)
  - $\pi_{\mathbf{x}}(\mathbf{z})$ is a proximity measure of input $\mathbf{z}$ to $\mathbf{x}$
  - $L(f, g, \pi_x)$ is a measure of how unfaithful g is in approximating f in a neighborhood given by $\pi_{\mathbf{x}}$

- Explanation:
  - $$g^* = \text{argmin}_{g \in G} L(f, g, \pi_{\mathbf{x}}) + \Omega(g)$$
  - How do we solve it? This is an optimization problem.
    - Use sampling (local exploration of points $\mathbf{z}$ in a neighborhood of $\mathbf{x}$, weighted by $\pi_{\mathbf{x}}$)

Interpretable ML: LIME Example

- Ribeiro, Marco Tulio, Sameer Singh, and Carlos Guestrin. "*Why should I trust you?: Explaining the predictions of any classifier*." KDD, 2016
  - SVM RBF classifier to distinguish "christianity" from "atheism" (94% testing accuracy)
    - Dataset: subset of the 20 newsgroup dataset



Explanation: the classifier uses irrelevant information (is right "by accident")

- E.g.: "Posting" appears in 22% of examples in the training data and in 99% of those in the class "Atheism"

What went wrong here?
How would you fix it?

Interpretable ML: LIME Example

- Ribeiro, Marco Tulio, Sameer Singh, and Carlos Guestrin. "*Why should I trust you?: Explaining the predictions of any classifier*." KDD, 2016
  - Google's pre-trained Inception neural network

(a) Original Image (b) Explaining *Electric guitar* (c) Explaining *Acoustic guitar* (d) Explaining *Labrador*

Figure 4: Explaining an image classification prediction made by Google's Inception neural network. The top 3 classes predicted are "Electric Guitar" ($p = 0.32$), "Acoustic guitar" ($p = 0.24$) and "Labrador" ($p = 0.21$)

Super-pixels explanations (with K = 10)

- Note: top predicted class is wrong

Interpretable ML: LIME Example

- Ribeiro, Marco Tulio, Sameer Singh, and Carlos Guestrin. "*Why should I trust you?: Explaining the predictions of any classifier*." KDD, 2016
  - Can explanations provide insights? (experiment with graduate students.)
    - Experiment: train a bad model to distinguish “husky” from “wolf”, then ask questions to subjects before and then again after they are show the explanation

(a) Husky classified as wolf (b) Explanation

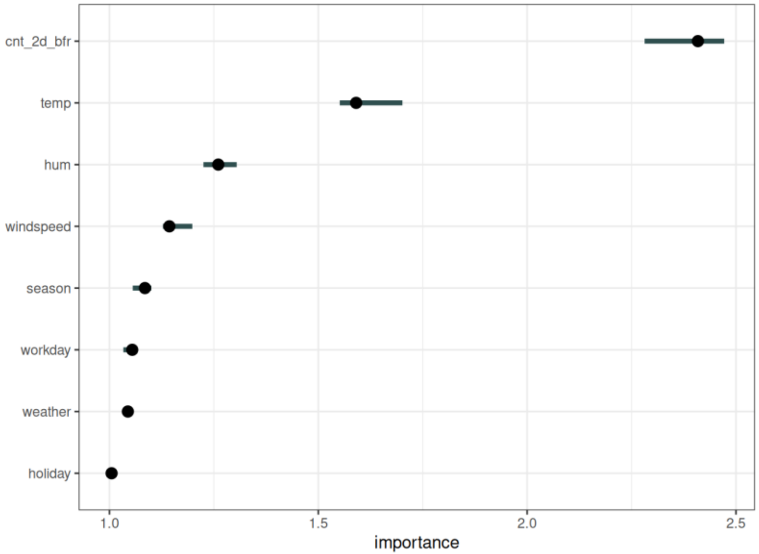
Figure 11: Raw data and explanation of a bad model’s prediction in the “Husky vs Wolf” task.

	Before	After
Trusted the bad model	10 out of 27	3 out of 27
Snow as a potential feature	12 out of 27	25 out of 27

Table 2: “Husky vs Wolf” experiment results.

Permutation Feature Importance (PFI)

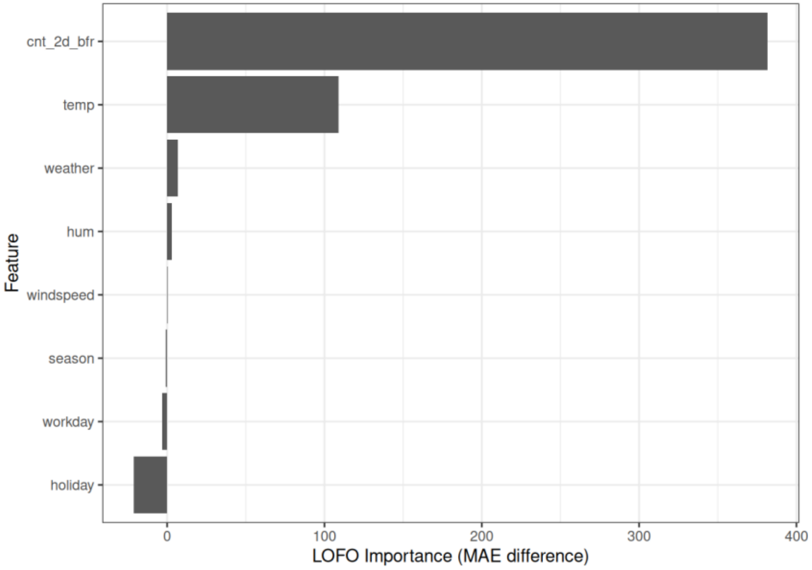
- Fisher, Rudin, and Dominici. "*All models are wrong, but many are useful: Learning a variable's importance by studying an entire class of prediction models simultaneously.*" JMLR 2019.
  - <u>Simple idea:</u> a feature is ***important*** if when we **shuffle** its values **model error increases**
    - It implies the model relies on this feature



source: https://christophm.github.io/interpretable-ml-book

Leave One Feature Out Importance (LOFO)

- If we want to know if a feature is important. Why not retrain the model without it?
  - **LOFO**: We do exactly this (for each feature) and compute the **error before** and after retraining
    - err_{orig}: error before retraining; err_{-j}: error after retraining without feature j
    - $LOFO_j = err_{-j} - err_{orig}$



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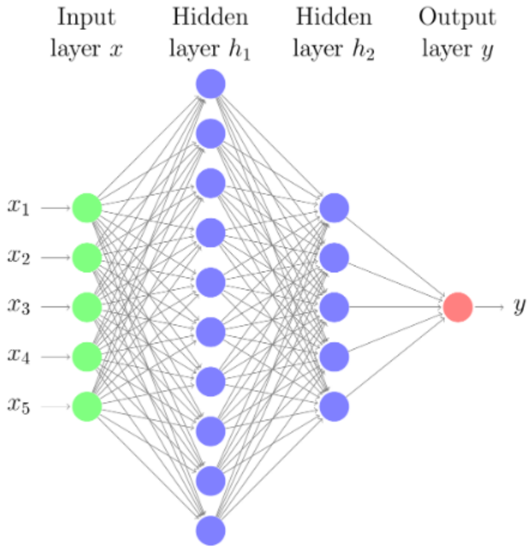
Surrogate / Proxy Models

- Proxy models techniques
  - Idea: create (train) a **global** *proxy model* that behaves similarly but is easier to explain
    - Linear proxy models
    - Decision trees
    - Rule extraction (e.g., if-then rules, MofN)
  - Note: LIME also uses proxy models but ***locally***
    - Here we are talking about training a ***new model*** to capture how the original model behaves
- Important:
  - Surrogate ***intrinsically interpretable*** models can be useful
  - But: we are **explaining the new model** (**not the data** and **not original model**)
    - In other words: this could provide the illusion of interpretability

Rule Extraction: DeepRED

- Zilke, Jan Ruben, Eneldo Loza Mencía, and Frederik Janssen. "*DeepRED–Rule extraction from deep neural networks*." International Conference on Discovery Science, 2016.




```
IF X1<0.5 AND X2>0.75 THEN OUT=1
IF X1>0.9 THEN OUT=1
IF X1>0.5 AND X1<0.9 AND X3>0.2 THEN OUT=1
IF X2>0.2 AND X3<0.5 AND X5<0.5 THEN OUT=1
IF X2>0.4 AND X3<0.7 THEN OUT=1
IF X2<0.2 THEN OUT=1
IF X4>0.8 THEN OUT=1
IF X3<0.7 AND X3>0.2 AND X4<0.3 THEN OUT=1
```

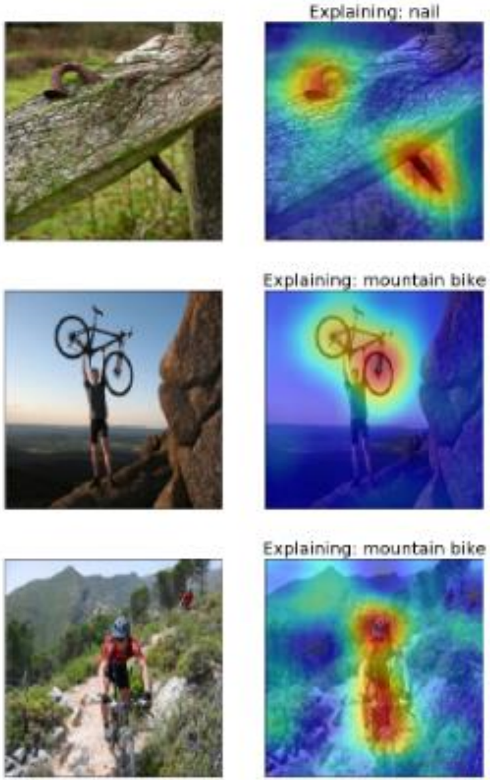
Interpretable ML: Saliency Maps

- Instead of training a proxy model, can we directly highlight the salient features?
  - Similar to what LIME is doing with super-pixels
  - We want to see the small portion of the input that is the most relevant
    - Note: sometimes called “attribution”

- How?
  - Simple solution:
    - Repeatedly occlude portions of the input
    - Create a map showing which part of the input actually influences the output

  - More efficient (& better):
    - Compute the saliency map directly by computing the input gradient (or other related quantities)
    - Example of techniques: Class Activation Mapping (CAM), Grad-CAM, Integrated gradients, etc.

Interpretable ML: Saliency Maps



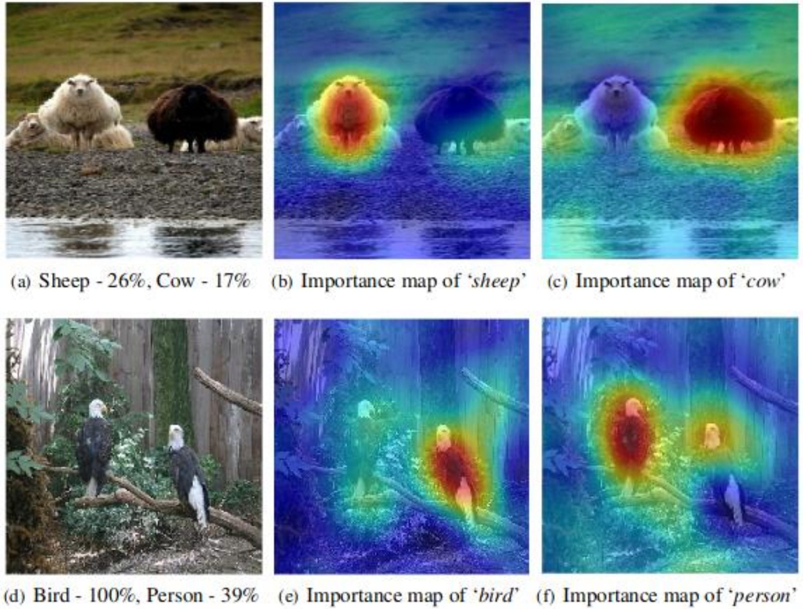


Figure 1: Our proposed RISE approach can explain why a black-box model (here, ResNet50) makes classification decisions by generating a pixel importance map for each decision (redder is more important). For the top image, it reveals that the model only recognizes the white sheep and confuses the black one with a cow; for the bottom image it confuses parts of birds with a person. (Images taken from the PASCAL VOC dataset.)

Source: Petsiuk, Das, and Saenko. "RISE: Randomized input sampling for explanation of black-box models." BMVC, 2018.

Next Time

- Wednesday (4/15): Lecture

- Upcoming:
  - **Homework 5** due **Friday 4/17**
  - **Project Deliverables** due **Wednesday 4/22** (no late penalty if submitted by **4/24**)
  - **Final Exam** on **Wednesday 4/22** at **10:40** (in class in **NRN 1020**)